

Responsible AI Report

Smart City Traffic Intelligence — Capstone Project

Stanley Neo

NUS School of Computing — AI/ML/DS Programme

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1. Purpose and Scope

This report documents the responsible-AI considerations for the Smart City Traffic Intelligence system: a machine-learning pipeline that forecasts hourly traffic volume and congestion risk on a US interstate highway (Metro Interstate dataset, 2012–2018, 40,575 cleaned hourly records) and converts those forecasts into recommendations for commuters and traffic operations staff. It covers data ethics, model limitations, fairness, transparency, accountability, and environmental impact, together with the safeguards implemented in the project.

2. Data Ethics and Privacy

The dataset contains only aggregated, anonymised sensor readings (hourly vehicle counts, weather measurements, calendar markers). It contains **no personally identifiable information** — no number plates, no individual journeys, no location traces of persons — so privacy risk is minimal. The data is public research data, used under its original terms for educational purposes.

Two representativeness caveats are explicitly acknowledged:

— **Single-corridor bias.** All records come from one highway station. Models trained on it describe this corridor's rhythm, not "traffic" in general.

— **Geographic and cultural transfer.** The data is from the US Midwest. Driving patterns, holiday calendars, and weather responses differ substantially in other cities (e.g. Singapore), so the models must not be deployed elsewhere without retraining and re-validation on local data.

3. Model Limitations

Limitation	Consequence	Mitigation in this project
Proxy ground truth. "Congestion" is defined as traffic volume $\geq$ the 75th percentile (4,952 vehicles/h); "high risk" is a constructed combination of congestion and severe weather. Neither is an observed incident label.	The classifier learns our definition, not real-world danger.	Definitions are documented in the code and reports; the high-risk model is framed as a screening aid, never an automated incident detector.
Class imbalance. Only 1.46% of hours are high-risk.	A naive model scores 98.5% accuracy by predicting "never".	Balanced class weights; evaluation by recall (0.95) and precision, not accuracy.

Historical $\neq$ future. Models assume 2012–2018 patterns persist.	Structural changes (roadworks, policy, pandemics) break forecasts.	Time-based train/test split; PSI drift monitoring with PASS/ALERT verdicts and a documented retraining trigger.
Observed weather, not forecast weather. Features use the weather recorded <i>during</i> the target hour.	Real deployment needs weather forecasts, adding error.	Limitation stated in reports; calendar features (which need no forecast) carry most of the predictive signal, keeping the system useful even with imperfect weather inputs.

4. Fairness and Bias

Traffic recommendations can redistribute congestion rather than reduce it: advising all commuters to shift to "quiet hours" could create new peaks and shift burden onto people whose jobs do not allow flexible schedules (shift workers, care-givers). The recommender therefore presents advice as one input among many and always pairs commuter advice with infrastructure-side actions (ramp metering, incident response) that do not depend on individual compliance.

Holiday behaviour is learned from only 53 holiday dates, and the model knows US federal holidays only; predictions for festive periods elsewhere would be systematically wrong. This is disclosed as a known bias rather than hidden in aggregate accuracy.

5. Transparency and Explainability

Every model in the system is accompanied by an explanation mechanism:

— **Global:** RandomForest feature importances and SHAP summary plots show that calendar features (hour-of-day, rush-hour flag, day-of-week) dominate predictions, with weather contributing small situational adjustments — consistent with the exploratory findings in Part 1.

— **Local:** SHAP values allow any single prediction to be decomposed into per-feature contributions, so an operator can audit why a specific hour was flagged.

— **Rules:** association rules (e.g. {Afternoon, Weekday} $\rightarrow$ High traffic, lift 2.86) give non-technical stakeholders an interpretable view of the same patterns.

— **Reproducibility:** all experiments are logged in MLflow (parameters, metrics, model artifacts), all code is version-controlled in Git, and every pipeline step writes to a persistent log file.

6. Accountability and Human Oversight

The system is decision-support, not decision-making. All outputs carry risk labels and historical context; no automated control action is taken. The monitoring script operationalises accountability: it compares incoming data against the training reference (PSI), re-scores model accuracy on new batches, and only triggers retraining when a data ALERT coincides with performance degradation — a deliberately conservative policy that avoids both silent failure and alarm fatigue. During development, the monitor flagged a genuine single-feature drift (cloud-cover distribution in 2018, PSI 0.267) while accuracy remained unchanged (MAE 241.2); the documented action was "log and investigate, no retrain", illustrating the framework working as intended.

7. Environmental and Compute Considerations

All models are deliberately small: the RandomForests train in under a minute on a CPU, and the LSTM (4,897 parameters) trains in seconds. No GPU or cloud compute is required, keeping the carbon footprint of experimentation negligible and the project reproducible on ordinary hardware.

8. Summary of Risks and Safeguards

Risk	Severity	Safeguard
Proxy labels mistaken for real incidents	Medium	Documentation; screening-aid framing
Silent performance decay over time	High	PSI + performance monitoring, PASS/ALERT, retraining trigger
Over-trust in a black-box score	Medium	SHAP explanations, association rules, MLflow audit trail
Inappropriate geographic transfer	Medium	Explicit scope statement; retraining requirement
Privacy	Low	Aggregated public data; no PII anywhere in the pipeline

Conclusion: within its stated scope — one corridor, historical patterns, decision support only — the system meets responsible-AI expectations for transparency, accountability, and proportionate safeguards. Deployment beyond that scope requires retraining, local validation, and a renewed fairness review.